

Error Analysis: Baseline vs Fine-tuned Model Comparison for NLI4CT

Performance Analysis on Gold Test Dataset

February 23, 2026

1 Introduction

This analysis compares the performance of the baseline model (Qwen) and the fine-tuned model (Qwen with LoRA) on the Natural Language Inference task for clinical trials (NLI4CT). The evaluation is performed on 500 examples from the Gold Test dataset (`Gold_test_formatted.jsonl`), distributed across two task types (Single and Comparison) and four sections (Eligibility, Intervention, Results, Adverse Events).

2 Global Results

Model	Accuracy	Improvement
Baseline	66.40% (0.6640)	–
Fine-tuned	74.20% (0.7420)	+7.80 points

Table 1: Global accuracy on the Gold test set (500 examples).

The fine-tuned model shows a clear global improvement of about 8 points over the baseline.

3 Analysis by Task Type

3.1 Accuracy by Type

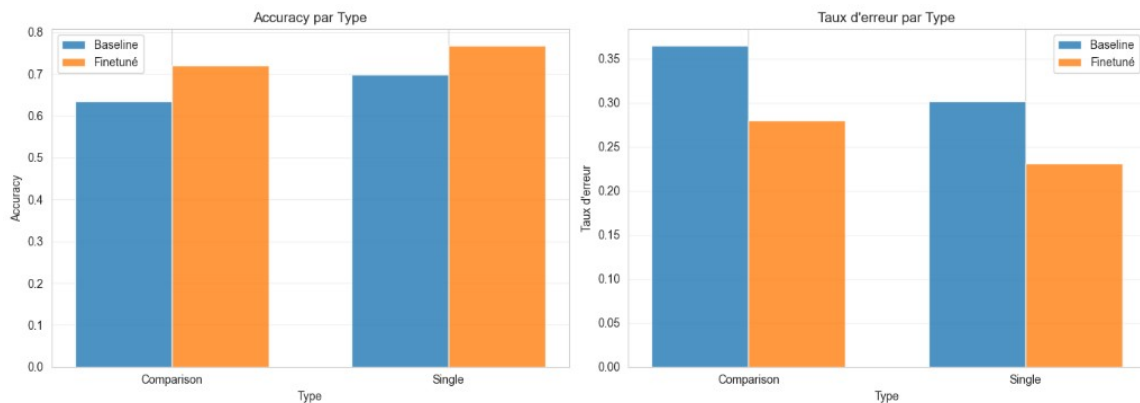


Figure 1: Accuracy and error rate by task type (Single vs Comparison).

Observations:

Type	Baseline	Fine-tuned
Comparison	0.635	0.720
Single	0.699	0.769

Table 2: Accuracy by task type.

- Fine-tuning improves accuracy by about **+0.085** for Comparison tasks and **+0.070** for Single tasks.
- Single tasks are easier than Comparison tasks for both models.
- The relative gain is larger for Comparison (+13% in accuracy) than for Single (+10%).

3.2 Error Rate by Type

Type	Baseline	Fine-tuned
Comparison	0.365	0.280
Single	0.301	0.231

Table 3: Error rate by task type.

4 Analysis by Section ID

4.1 Accuracy by Section

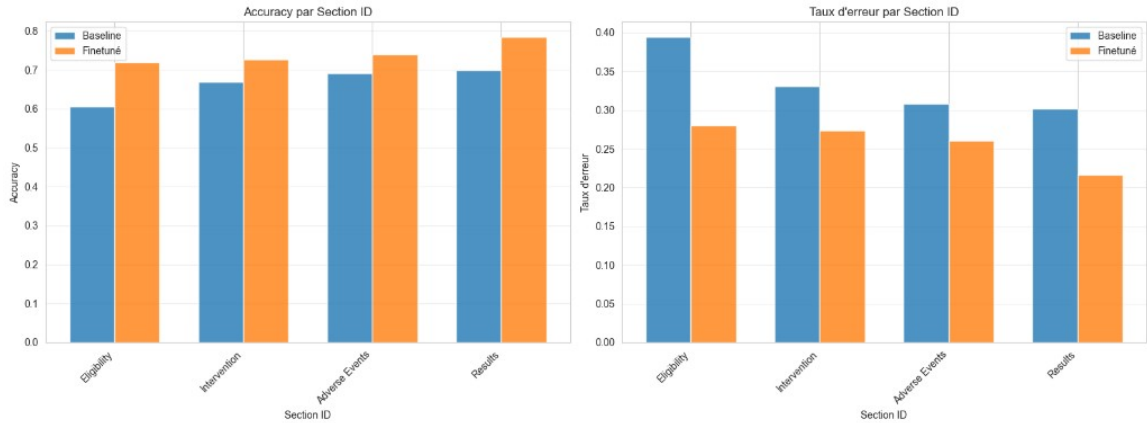


Figure 2: Accuracy and error rate by section (Eligibility, Intervention, Adverse Events, Results).

Observations:

- **Adverse Events:** Best section for both models; fine-tuned reaches 79.2%.
- **Eligibility:** Hardest section (62.1% baseline, 68.9% fine-tuned).
- Largest gains: Intervention (+0.106) and Results (+0.085).

Section ID	Baseline	Fine-tuned
Eligibility	0.621	0.689
Intervention	0.641	0.747
Results	0.660	0.745
Adverse Events	0.717	0.792

Table 4: Accuracy by section ID.

Section ID	Baseline	Fine-tuned
Eligibility	0.379	0.311
Intervention	0.359	0.254
Results	0.340	0.255
Adverse Events	0.283	0.208

Table 5: Error rate by section ID.

4.2 Error Rate by Section

5 Cross-Analysis: Type $\times$ Section

5.1 Baseline

- **Best:** Single / Adverse Events (0.729), Single / Results (0.730).
- **Worst:** Comparison / Eligibility (0.580), Comparison / Results (0.651).
- Comparison tasks are more variable across sections than Single.

5.2 Fine-tuned

- **Best:** Single / Eligibility (0.818), Comparison / Adverse Events (0.806).
- **Worst:** Comparison / Eligibility (0.671), but still above baseline.
- Performance is more homogeneous; large gains on previously weak cells (e.g. Comparison + Eligibility, Comparison + Intervention).

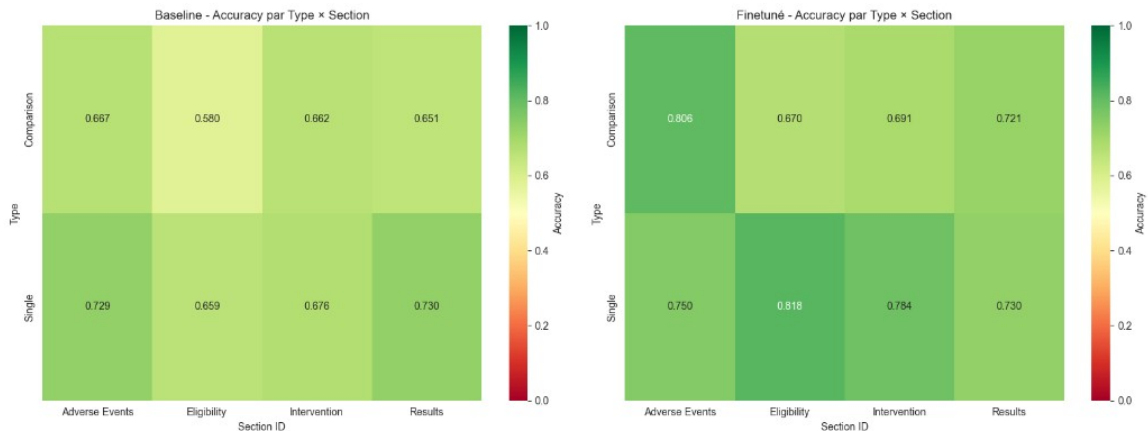


Figure 3: Baseline (left) and Fine-tuned (right) accuracy heatmaps by Type $\times$ Section.

5.3 Notable improvements

Combination	Baseline	Fine-tuned
Comparison + Eligibility	0.580	0.671 (+0.091)
Comparison + Intervention	0.662	0.691 (+0.029)
Single + Eligibility	0.659	0.818 (+0.159)
Single + Intervention	0.676	0.784 (+0.108)

Table 6: Selected Type $\times$ Section improvements.

6 Regression Analysis

6.1 Regression distribution

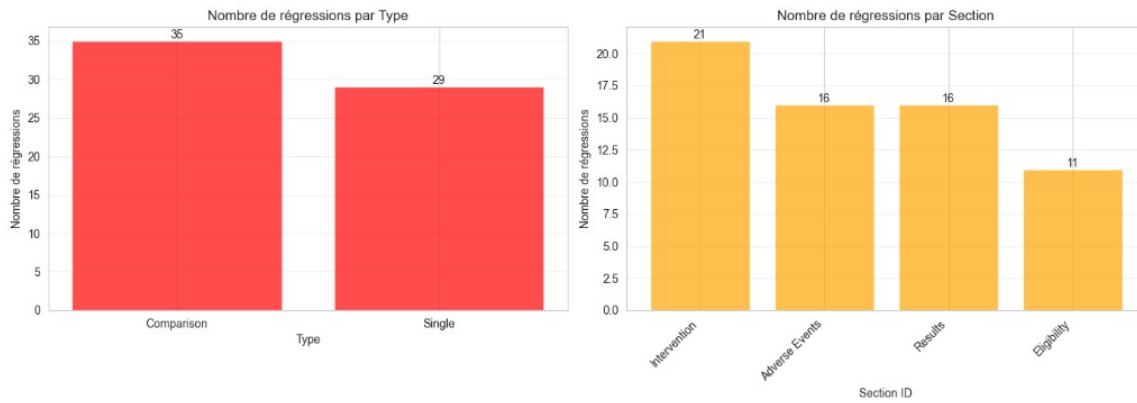


Figure 4: Number of regressions (baseline correct, fine-tuned incorrect) by Type and by Section.

Despite global improvements, fine-tuning introduces regressions (cases where the baseline was correct but the fine-tuned model is wrong):

Category	Number of regressions
By Type:	
Comparison	35
Single	29
By Section:	
Intervention	21
Adverse Events	16
Results	16
Eligibility	11
Total	64

Table 7: Regression distribution (64 regressions total, 12.8% of 500 examples).

Observations:

- **64 regressions** out of 500 examples (12.8%).
- Slightly more regressions for **Comparison** (35) than for Single (29).

- By section: most regressions in **Intervention** (21), then Adverse Events and Results (16 each), then Eligibility (11).

6.2 Interpretation

Regressions (baseline correct, fine-tuned incorrect) can be explained by several factors:

1. **Overfitting:** Fine-tuning may overfit some patterns from the training set, leading to errors on cases that the baseline handled correctly.
2. **Baseline “Contradiction” bias:** The baseline often predicts “Contradiction”; on some examples it is therefore correct “by chance” without necessarily reasoning well. When the fine-tuned model predicts differently (often more Entailment after training), it can be wrong on those cases while the baseline, by systematically predicting Contradiction, had given the right answer. Part of the regressions thus reflects this baseline bias rather than a pure degradation of the fine-tuned model.
3. **Comparison** tasks are more sensitive to regressions, as they require reasoning over two trials.
4. Overall, the effect remains clearly positive (about +8 points accuracy) despite these regressions.

7 Examples by Category

The examples below illustrate the four possible situations (true label, baseline prediction, fine-tuned prediction). Statements (hypothesis/statement) are truncated when necessary.

7.1 Baseline Wrong, Fine-tuned Correct (Improvement)

Example 1.

- **Statement:** “The secondary trial requires patients to be over a certain age, but the primary trial does not specify an age range for participation.”
- True label: **Entailment**. Baseline: **Contradiction**. Fine-tuned: **Entailment**.

Example 2.

- **Statement:** “Patients with histologically confirmed stage 4 breast adenocarcinoma that is either ER positive, PR positive, or HER2/neu negative are eligible for both the secondary trial and the primary trial.”
- True label: **Contradiction**. Baseline: **Entailment**. Fine-tuned: **Contradiction**.

7.2 Baseline Correct, Fine-tuned Wrong (Regression)

Example 1.

- **Statement:** “There are no conditions on mental health, bodyweight, size of tumours, Karnofsky/ECOG score or Estimated life expectancy that need to be met in order to be eligible for the primary trial.”
- True label: **Entailment**. Baseline: **Entailment**. Fine-tuned: **Contradiction**.

Example 2.

- **Statement:** “A patient that has received an organ transplant within the last month, and is still bedridden, would be excluded from the primary trial but may be eligible for the secondary trial.”
- True label: **Contradiction**. Baseline: **Contradiction**. Fine-tuned: **Entailment**.

7.3 Both Models Wrong (Common Errors)

Example 1.

- **Statement:** “The primary trial is evaluating the number of patients who suffer treatment-related adverse events, whereas the secondary trial measures the median sleep time of its patients in minutes.”
- True label: **Entailment**. Baseline: **Contradiction**. Fine-tuned: **Contradiction**.

Example 2.

- **Statement:** “Every participant in the primary trial receives the same intervention, regardless of whether or not they have cancer.”
- True label: **Entailment**. Baseline: **Contradiction**. Fine-tuned: **Contradiction**.

7.4 Both Models Correct

Example 1.

- **Statement:** “Women suffering from both claustrophobia and IBS are not eligible for either the primary trial or the secondary trial.”
- True label: **Contradiction**. Baseline: **Contradiction**. Fine-tuned: **Contradiction**.

Example 2.

- **Statement:** “Patients currently taking part in the secondary trial are not eligible to enroll in the primary trial at the same time.”
- True label: **Entailment**. Baseline: **Entailment**. Fine-tuned: **Entailment**.

8 Conclusions

8.1 Positive Aspects

- **Clear global gain:** +7.8 points accuracy (66.4% $\rightarrow$ 74.2%).
- **Consistent gains by type:** both Single and Comparison improve.
- **Consistent gains by section:** all four sections improve, with largest gains on Intervention and Results.
- **Cross-analysis:** the fine-tuned model improves the hardest Type $\times$ Section combinations (e.g. Comparison + Eligibility, Single + Eligibility).

8.2 Points of Concern

- **Regressions:** 12.8% of examples show regressions and deserve targeted analysis.
- **Comparison tasks:** more regressions (35 vs 29) and higher error rate than Single even after fine-tuning.
- **Eligibility:** remains the most difficult section (lowest accuracy).

8.3 Recommendations

1. Inspect the 64 regression cases to identify recurring failure patterns.
2. Enrich training with more Comparison examples, especially for Eligibility.
3. Consider hyperparameter tuning (e.g. lower learning rate, more epochs) to reduce regressions.

9 Appendix: Methodology

9.1 Dataset

- **Gold Test:** 500 examples from `Gold_test_formatted.jsonl`.
- **Types:** Single (one trial) and Comparison (two trials).
- **Sections:** Eligibility, Intervention, Results, Adverse Events (from `Gold_test.json`).

9.2 Models

- **Baseline:** Base model (e.g. Qwen).
- **Fine-tuned:** Same base with LoRA on NLI4CT.

9.3 Metrics

- **Accuracy:** Proportion of correct predictions.
- **Error rate:** 1 - Accuracy.
- **Regressions:** Examples where baseline is correct and fine-tuned is incorrect.